UAV Autopilot - QUT open source version

Wednesday, 12 October 2022 1:41 PM

Project Summary

To create QUT's own open source version of Autopilot Hardware that is compatible with PX4 or ArduPilot Firmware

Project Owner: Seth Arekhajak (WIL Student)
Project Owner: Kai-Cherng (Wesley) (WIL Student - past)
Project Supervisor: Brendon
Project Manager: Brendon

Technical: Steve
Technical:

Software: Altium

Start: 25/04/2022 (Wesley) 22/05/2023 (Seth) 24/06/2024 (Cory) 1/12/2025(Joshua)
End: 20/10/2022 (Wesley) 7/06/2023 (Seth) 15/08/2024 (Cory)

Folder :: U:\SEF2.0\TechnicalServices\EEI\Projects\QUT_UAV_Autopilot

Location: S913 S - Block
RA:
Account:

The unmanned aerial vehicle (UAV) hardware landscape is currently bifurcated between high-performance, proprietary systems and aging open-source reference designs. While the modern "Pixhawk 6X" (FMUv6X) offers superior processing via the STM32H7 microcontroller, its schematics are often restricted under non-commercial or proprietary licenses, limiting the ability of independent engineering teams to manufacture, modify, or brand the hardware without incurring significant licensing fees or legal ambiguity.

Conversely, the legacy FMUv3/4 architecture—best known as the "Pixhawk 2" or "The Cube"—remains the most robust, fully open-source hardware definition available under the Creative Commons Attribution-Share A like 3.0 license.

The objective is to design a "Neo-FMUv3" or "FMUv4 "autopilot: a flight controller that retains the legally accessible and firmware-compatible architecture of the Pixhawk 2.
The Purpose is to have access to an Autopilot that can be custom ordered whenever it is required instead of waiting for stock of off the shelf Autopilots.

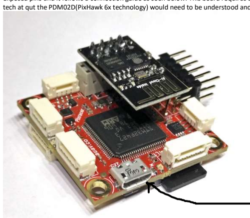
There are 2 approaches,

FMUv3: Which has a wide selection of Open source designs, however it is 10 years old so most sensor components are no longer in production now, Alternatives need to be found and designs need to be modified to fit new pinouts (difficult). See research word doc in folder

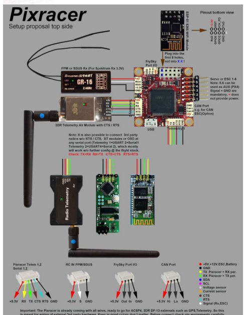
Pixhawk-https://github.com/pixhawk/Hardware.git
Cube-https://github.com/proficnc/The-Cube.git
3Dr-https://github.com/3drobotics/Pixhawk_OS_Hardware.git

Links to Open source schematics, This type of AutoPilot has 4 boards-> IMU board which holds sensors and is suspended in foam above FMU, FMU board which has main chips and 2 connections(1 to IMU and 1 to PSM), 1 PSM board which connects to the FMU board. Finally a Carrier board which connects via a 80 pin connector(the big black board). The schematics for this can be found on Cube website however it is NOT Open source(the files are there but if you use them its technically not legal)

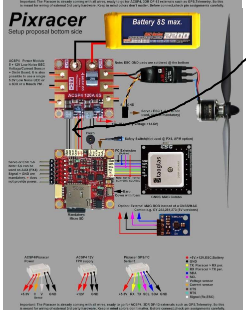
FMUv4: While the opensource hardware stopped after FMUv3, some products still remained opensource including the PixRacer. This Autopilot uses the FMUv4 standard which has less out of date components but less opensource information. This board is a lot more compact and removes the need for 4 different boards. The PWM is attached via the side exposed pins and it follows a connection guide as seen below. This board requires a PDB(power distribution board) chip as seen in the below images. To make this work with current tech at qut the PDM02D(PixHawk 6x technology) would need to be understood and adapted into this new design.

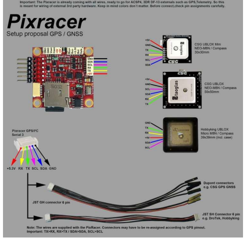


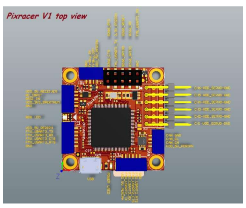
usb c?



REPLACE with PDM02D?





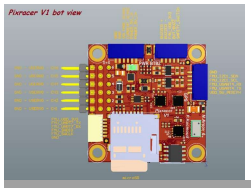


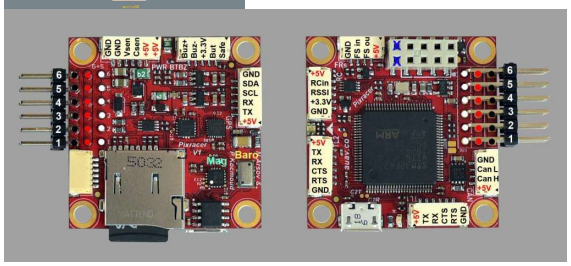
Timeframe, Key Dates

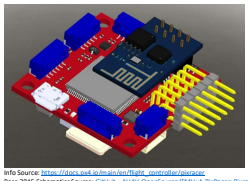
Date	Key Task	Person
25/04/2022	Project Start	Wes
27/04/2022 - 04/05/2022	Research	Wes
06/05/2022 - 20/05/2022	Schematic Drawing	Wes
23/05/2022 - 03/06/2022	Schematic Drawing Component Search	Wes
08/06/2022 - 22/06/2022	Component Search	Wes
28/06/2022 - 08/07/2022	Schematic Drawing Component Search	Wes
12/07/2022 - 26/07/2022	Board Layout Design V1 PCB Order	Wes
28/07/2022 - 11/08/2022	PCB Order Review Component Search / Check Component Order	Wes
16/08/2022 - 24/08/2022	Component Stock Review Component Order	Wes
25/08/2022 - 13/09/2022	Soldering Preparation	Wes
14/09/2022 - 27/09/2022	Populating PCB V1 Soldering Using Reflow Testing Sensors	Wes
28/09/2022 - 11/10/2022	Sensor Testing Redo Some Soldering	Wes
12/19/2022 - 20/10/2022	Sensor Testing Handover Preparation	Wes
20/10/2022	Handover	Wes
29/05/2023	New sensors ordered for a replacement autopilot	Seth
7/06/2023	Replacement sensors arrived Handover	Seth
24/06/2024	Research	Cory
1/07/2024	Found issues with old PCB and made changes to the schematic on Altium.	Cory
10/07/2024	Began testing every major flight controller sensor through writing Arduino programs to view sensor data.	Cory
17/07/2024	Made more changes to PCB schematic as well as adding extra test points for key components.	Cory
29/08/2024	Ensured the PCB followed design rules for high frequency components and prepared Gerber files for new PCB order.	Cory
15/08/2024	Changes made to old PCB schematic, ready to order new PCB. Handover.	Cory
1/12	Read Handover and understand designs and current progress. Refactored the OneNote.	Josh

Component Code	Description	Schematic	Provider	Provider Compoent Code	Link
CDBU70	Schottky Diodes & Rectifiers Schottky Barr Diode 70mA, 70V	(D1)	Mouser	750-CDBU70	https://au.mouser.com/ProductDetail/Comchip
TSW-106-08-F-T-RA	Headers & Wire Housings Classic PCB Header Strips, 0.100 pitch	(J10, J11, J12)	Mouser	200-TSW10608FTRA	https://au.mouser.com/ProductDetail/Samtec/
C1608X7S1C225K080AE	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0603 16VDC 2.2uF 10% X7S	C1, C7, C12, C13, C14, C17, C20, C22, C27, C36, C37, C38, C40, C47	Mouser	810-C1608X7S1C225K08	https://au.mouser.com/ProductDetail/TDK/C16
CGA3E3X7T0G106M080AB	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0603 4VDC 10uF 20% X7T AEC-Q200 EPOXY ONLY	C15, C29, C43	Mouser	810-CGA3E3X7T0G106M0	https://au.mouser.com/ProductDetail/TDK/CGA
GRM31CC80G227ME11L	Multilayer Ceramic Capacitors MLCC - SMD/SMT 220UF 4V 20% 1206	C16	Mouser	81-GRM31CC80G227ME1L	https://au.mouser.com/ProductDetail/Murata-
C0603C104K5RAC3121	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50V 0.1uF X7R 0603 10% AEC-Q200	C2, C5, C6, C8, C9, C10, C11, C18, C19, C21, C23, C25, C28, C30, C31, C32, C35, C45, C46, C48	Mouser	80-C603C104K5RAC3121	https://au.mouser.com/ProductDetail/KEMET/
EMF107B7474KAHT	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0603 16VDC 0.47uF 10% X7R AEC-Q200	C24	Mouser	963-EMF107B7474KAHT	https://au.mouser.com/ProductDetail/Taiyo-Yu
W2H13C1048AT1F	Feed Through Capacitors 25V 0.1uF X7R	C26, C39	Mouser	581-W2H13C1048AT1F	https://au.mouser.com/ProductDetail/Kyocera
C0603C809C5GACAUTO	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50V 8pF C0G 0603 0.25pF AEC-Q200	C3, C4	Mouser	80-C0603C809C5GAUTO	https://au.mouser.com/ProductDetail/KEMET/
GCM188R71H103KA37J	Multilayer Ceramic Capacitors MLCC - SMD/SMT PLEASE SEE MURATA'S SUGGESTED ALTERNATE GCM155R71H103KA55D	C33, C34	Mouser	81-GCM188R71H103KA7J	https://au.mouser.com/ProductDetail/Murata-
GRM31CR61A107MEA8K	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100UF 10V 20% 3216	C41, C42	Mouser	81-GRM31CR61A107MA8K	https://au.mouser.com/ProductDetail/Murata-
GRM31CR61A107ME05L	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100UF 10V 20% 1206	C41, C42	Mouser	81-GRM31CR61A107ME5L	https://au.mouser.com/ProductDetail/Murata-
JMK107ABJ106KAHT	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0603 6.3VDC 10uF 10% X5R AEC-Q200	C44	Mouser	963-JMK107ABJ106KAHT	https://au.mouser.com/ProductDetail/Taiyo-Yu
PMEG2020CPASX	Schottky Diodes & Rectifiers 20V 2A dual MEGA Schottky	D2	Mouser	771-PMEG2020CPASX	https://au.mouser.com/ProductDetail/Nexperia
MMSZ5232BT1G	Zener Diodes 5.6V 500mW	D3	Mouser	863-MMSZ5232BT1G	https://au.mouser.com/ProductDetail/onsemi/
PMEG1030EJ,115	Schottky Diodes & Rectifiers SCHOTTKY 10V 3A	D4A	Mouser	771-PMEG1030EJ-T/R	https://au.mouser.com/ProductDetail/Nexperia
PMEG1030EH,115	Schottky Diodes & Rectifiers PMEG1030EH/SOD123F/SOD2	D4B	Mouser	771-PMEG1030EH-T/R	https://au.mouser.com/ProductDetail/Nexperia
1N4448X-TP	Diodes - General Purpose, Power, Switching 250mA 75V 500mA 250mA 150mW 75Vbr	D5	Mouser	833-1N4448X-TP	https://au.mouser.com/ProductDetail/Micro-C
0ZCJ0200FF2C	Resettable Fuses - PPTC Fuse	F1	Mouser	530-0ZCJ0200FF2C	https://au.mouser.com/ProductDetail/Bel-Fuse
0ZCK0035FF2G	Resettable Fuses - PPTC	F2	Mouser	530-0ZCK0035FF2G	https://au.mouser.com/ProductDetail/Bel-Fuse
0ZCJ0100FF2E	Resettable Fuses - PPTC	F3	Mouser	530-0ZCJ0100FF2E	https://au.mouser.com/ProductDetail/Bel-Fuse
STM32F427VIT6		IC1			
74LVC2G86DC,125	Logic Gates DUAL 2-IN EX-OR GATE	IC10	Mouser	771-LVC2G86DC125	https://au.mouser.com/ProductDetail/Nexperia
AP7365-33WG-7	DiodesZetex AP7365-33WG-7, 1 Low Dropout Voltage, Voltage Regulator 600mA, 3.3 V 5-Pin, SOT-25	IC11	RS Components	921-0619	https://my.rs-online.com/web/p/voltage-regula
MCP131T-300E/TT	Supervisory Circuits Open Drain Low	IC2	Mouser	579-MCP131T-300E/TT	https://au.mouser.com/ProductDetail/Microch
FM25V02A-G	NVRAM NVRAM FRAM Memory Serial	IC3	Mouser	727-FM25V02A-G	https://au.mouser.com/ProductDetail/Cypress-
ICM-20948	IMU ACCEL/GYRO/COMPI2C/SPI 24QFN	IC4	Win Source Electronics	846576-ICM-20948	https://www.win-source.net/tdk-invensense-ic
EV_ICM-20602	Multiple Function Sensor Development Tools ICM-20602	IC5	Mouser	410-EV_ICM-20602	https://au.mouser.com/ProductDetail/TDK-Inve
ICM-20608-G	IMU ACCEL/GYRO I2C/SPI 16LGA	IC5	Digi-Key	1428-1052-1-ND	https://www.digikey.com.au/en/products/deta &utm_id=go_cmp-17853854068_adg-_ad-__de
MS561101BA03-50	Board Mount Pressure Sensors MS5611-BT M-CAPS 3X5X1MM T&R	IC6	Mouser	824-MS561101BA03-50	https://au.mouser.com/ProductDetail/Measure
NCP163ASN180T1G	LDO Voltage Regulators LDO Reg, Ultra-Low Noise High PSRR 1.8V	IC7	Mouser	863-NCP163ASN180T1G	https://au.mouser.com/ProductDetail/onsemi/
TJA1051TK/3,118	CAN Interface IC HIGH-SPEED CAN TRANSCEIVER	IC8	Mouser	771-TJA1051TK/3118	https://au.mouser.com/ProductDetail/NXP-Sem
TJA1051TK/3,118		IC8	Element14	2776121	https://au.element14.com/nxp/tja1051tk-3-11
74LVC1G86SE-7	Logic Gates LOGICLVCXOR GATE1	IC9	Mouser	621-74LVC1G86SE-7	https://au.mouser.com/ProductDetail/Diodes-I
SM06B-SRSS-TB (LF)(SN)	PCB connector SH base These are not the standard JST type connectors we use on most of our boards, but are actually the much smaller JST-SH type.	J1	RS Components	546-8827	https://au.rs-online.com/web/p/pcb-headers/5
47346-0001	USB Connectors MICRO USB B RECPT BTTM MOUNT ASSY	J2	Mouser	538-47346-0001	https://au.mouser.com/ProductDetail/Molex/4
SM06B-GHS-TB(LF)(SN)	1.25mm Wire to Board GH Series Side Entry Headers	J3, J7, J8, J9	RS Components	752-1800	https://au.rs-online.com/web/p/pcb-headers/7
PJS008-2120-0	Memory Card Connectors STD MNT MICRO SD PUSH-PUSH, SMT CONN	J4	Mouser	945-PJS008-2120-0	https://au.mouser.com/ProductDetail/Yamaich
SM04B-GHS-TB(LF)(SN)	1.25mm Wire to Board GH Series Side Entry Headers	J5, J14	RS Components	752-1797P	https://au.rs-online.com/web/p/pcb-headers/7 &gclid=EAIaIQobChMI1ITwjq7-9wIVFBwrCh09S
SM05B-GHS-TB(LF)(SN)	1.25mm Wire to Board GH Series Side Entry Headers	J6, J13	RS Components	752-1807P	https://au.rs-online.com/web/p/pcb-headers/7 &gclid=EAIaIQobChMI7JgB-rP-9wIVtZNmAh2j4
SSM-105-L-DV	Headers & Wire Housings Surface Mount PCB Socket Strips, 0.100 pitch	JP1	Mouser	200-SSM105LDV	https://au.mouser.com/ProductDetail/Samtec/
APG1608ZGCK	Standard LEDs - SMD 1.6X0.8MM	LED1	Mouser	604-APG1608ZGCK	https://au.mouser.com/ProductDetail/Kingbrig
LTST-C19HE1WT	Standard LEDs - SMD LED Red 615nm Grn 525nm/Blu 470nm	LED2	Mouser	859-LTST-C19HE1WT	https://au.mouser.com/ProductDetail/Lite-On/
XP151A12A2MR-G	MOSFET Power MOSFET, 20V, 1A, N-Type, SOT-23	Q1	Mouser	865-XP151A12A2MR-G	https://au.mouser.com/ProductDetail/Torex-Se
BSS209PW H6327	MOSFET P-Ch -20V -630mA SOT-323-3	Q2	Mouser	726-BSS209PWH6327	https://au.mouser.com/ProductDetail/Infineon
CMP0603-FX-1002ELF	Thick Film Resistors - SMD ResHighPower 0603 10k 1% 1/4W TC100	R1, R2, R12, R13, R21, R42, R48, R49, R53	Mouser	652-CMP0603-FX-1002L	https://au.mouser.com/ProductDetail/Bourns/
CMP0603AFX-1004ELF	Thick Film Resistors - SMD ResHighPowerA 0603 1M 1% 1/4W TC100	R10, R11	Mouser	652-CMP0603AFX-1004L	https://au.mouser.com/ProductDetail/Bourns/
ERJ-UP3D1003V	Thick Film Resistors - SMD AEC-Q200	R14, R15, R16, R17, R18	Mouser	667-ERJ-UP3D1003V	https://au.mouser.com/ProductDetail/Panason
SDR03EZPD1200	Thick Film Resistors - SMD	R19	Mouser	755-SDR03EZPD1200	https://au.mouser.com/ProductDetail/ROHM-S
CRCW0603220RJNEAC	Thick Film Resistors - SMD 1/10Watt 220ohms 5% Commercial Use	R3, R20, R24, R25, R26, R27, R28, R29, R30, R31, R34, R35, R36, R37, R38, R39, R44, R45, R46	Mouser	71-CRCW0603220RJNEAC	https://au.mouser.com/ProductDetail/Vishay-D
ERA-3VEB1501V	Thin Film Resistors - SMD 0603 0.1% 1.5Kohm 25ppm SMD	R32, R33, R40	Mouser	667-ERA-3VEB1501V	https://au.mouser.com/ProductDetail/Panason
RCC06031K00FKEA	Thick Film Resistors - SMD 1/5W 1Kohms 1% 100ppm	R4, R8, R9, R22, R23	Mouser	71-RCC06031K00FKEA	https://au.mouser.com/ProductDetail/Vishay-D
SDR03EZPD68R0	Thick Film Resistors - SMD	R41	Mouser	755-SDR03EZPD68R0	https://au.mouser.com/ProductDetail/ROHM-S
CHP0603AFX-4700ELF	Thick Film Resistors - SMD ResHighPowerA 0603 470R 1% 1/3W TC100	R43	Mouser	652-CHP0603AFX-4700L	https://au.mouser.com/ProductDetail/Bourns/
ERJ-UP3F10R0V	Thick Film Resistors - SMD 0603 0.25W 1% 10ohm AEC-Q200	R47	Mouser	667-ERJ-UP3F10R0V	https://au.mouser.com/ProductDetail/Panason
SDR03EZPF1000	Thick Film Resistors - SMD 0603 100ohm 1% Anti Surge AEC-Q200	R50, R51, R52	Mouser	755-SDR03EZPF1000	https://au.mouser.com/ProductDetail/ROHM-S
SDR03EZPD22R0	Thick Film Resistors - SMD	R6, R7	Mouser	755-SDR03EZPD22R0	https://au.mouser.com/ProductDetail/ROHM-S
LP5907QMFX-3.3Q1	LDO Voltage Regulators Auto 250mA,Ultra- Low Noise LDO Reg	U1, U2	Mouser	595-LP5907QMFX-3.3Q1	https://au.mouser.com/ProductDetail/Texas-In
LIS3MDLTR	Board Mount Hall Effect/Magnetic Sensors Ultra Lo-Pwr Hi Perf 3-Axis 1.9-3.6V	U3	Mouser	511-LIS3MDLTR	https://au.mouser.com/ProductDetail/STMicro
FA-20H 24.0000MF20X-K3	Crystals 24MHz 10ppm 10pF -40C +85C	Y1	Mouser	732-FA20H-24F20X-K3	https://au.mouser.com/ProductDetail/Epson-T
S 1 2 missing IC22 missing (ICM42688)					

	Cost/Unit	Quantity	Subtotal	Note	Ordered	Arrived	Stocktake 12/2025
-Technology/CDBU70?qs=J1R1puj4KiHu6WmHr9F3%252Bw%3D%3D	0.572	1	0.572	Not The same one as schematic, double check if compatible	YES	YES	YES
TSW-106-08-F-T-RA?qs=%252BZP6%2F%252BtExtAMBmbTUZgOoQ%3D%3D	$3.68	1	$3.68	(J10, J11, J12) All in one right angle	YES	YES	YES
608X7S1C225K080AE?qs=sGAEpiMZZMsh%252B1woXyUXj4o8uGOj6yOa%2F2JNARaH%252BUU%3D	$0.48	14	$6.78	2.2uF, 603 (in)	YES	YES	YES
A3E3X7T0G106M080AB?qs=sGAEpiMZZMsh%252B1woXyUXjzCAKOm63x3oUafVoeFPjqg%3D	$0.75	3	$2.24	10uF, 603 (in)	YES	YES	YES
Electronics/GRM31CC80G227ME11L?qs=sGAEpiMZZMsh%252B1woXyUXj%2FxCjkItOqpdtJ17yM6JR6A%3D	$1.72	1	$1.72	220uF 4V, 1206 (in)	YES	YES	YES
C0603C104K5RAC3121?qs=oBCMsStVSxe1EkGyoyETtg%3D%3D	$0.19	20	$3.82	0.1uF (100nF), 603 (in)	YES	YES	YES
uden/EMF107B7474KAHT?qs=sGAEpiMZZMsh%252B1woXyUXj30ZYomYlxpXbZmTEzBsZ7s%3D	$0.24	1	$0.24	470nF, 603	YES	YES	YES
-AVX/W2H13C1048AT1F?qs=yRKuMHbOG1onjNKOzcED5w%3D%3D	$0.72	2	$1.44	Filter Cap.	YES	YES	YES
C0603C809C5GACAUTO?qs=u4fy%2FsgLU9N7MIfzP7pNhA%3D%3D	$0.70	2	$1.41	8.0pF, 603 (in)	YES	YES	YES
Electronics/GCM188R71H103KA37J?qs=N3Kl9KD794T3%2FxlXdF0Axg%3D%3D	$0.15	2	$0.29	0.01uF (10nF) , 603 (in)	YES	YES	YES
Electronics/GRM31CR61A107MEA8K?qs=hd1VzrDQEGiKPwOlEhCnYw%3D%3D	$1.03	2	$2.06	100uF, 10V, 1206 (in) (On Backorder, Was in stock... 27/10/22)	BACKORDER	NO	YES
Electronics/GRM31CR61A107ME05L?qs=chTDxNqvsylSo93CzdbYkQ%3D%3D	$1.01	2	$2.02	100uF, 10V, 1206 (in) ^^Same as above	YES	YES	YES
uden/JMK107ABJ106KAHT?qs=UXgszm6BlbGVg%252B%252Bo5%252BNvhQ%3D%3D	$0.28	1	$0.28	10uF, 6.4V, 603 (in)	YES	YES	YES
a/PMEG2020CPASX?qs=K2OaAKmNuLimeK0EUCzHKQ%3D%3D	$0.76	1	$0.76		YES	YES	YES
MMSZ5232BT1G?qs=P4IOph%252Bbot%2FernOvVDK3lg%3D%3D	$0.38	1	$0.38	5.6V, SOT-23-3	YES	YES	YES
a/PMEG1030EJ115?qs=LOCUfHb8d9vmk7P%2FuYbs5A%3D%3D	$0.60	1	$0.60	Was in stock, Now on Backorder 3/10/22	BACKORDER	NO	YES
a/PMEG1030EH115?qs=LOCUfHb8d9uVcQd0kSivzA%3D%3D	$0.63	1	$0.63	^^ (Different Pacakge)	YES	YES	YES
ommercial-Components-MCC/1N4448X-TP?qs=0RDYWt4CO4fFI8HfN%2F4aEA%3D%3D	$0.43	1	$0.43		YES	YES	YES
/0ZCJ0200FF2C?qs=SRYZG9HaIQ2c%2F4nW1aZeIg%3D%3D	$0.45	1	$0.45	2A RST Fuse, 1206 (in)	YES	YES	YES
/0ZCK0035FF2G?qs=SRYZG9HaIQ1hwWyCkUgYKw%3D%3D	$0.21	1	$0.21	0.3A RST Fuse, 0805 (in)	YES	YES	YES
/0ZCJ0100FF2E?qs=SRYZG9HaIQ16trCtSE8WyA%3D%3D	$0.29	1	$0.29	1A RST Fuse, 1206 (in)	YES	YES	YES
		1	0	Ordered from AliExpress (fingercross will work)	YES	YES	YES
a/74LVC2G86DC125?qs=me8TqzrmIYXbZy3sMRjHUw%3D%3D	$1.01	1	$1.01		YES	YES	YES
ators/9210619	$0.82	1	$0.82	Min QTY Purchase 30	YES	YES	YES
ip-Technology/MCP131T-300E-TT?qs=oKK8NaWdAJuugCcbQvjtFw%3D%3D	$1.07	1	$1.07		YES	YES	YES
Semiconductor/FM25V02A-G?qs=sGAEpiMZZMutXGli8Ay4kEPfvY4JW7cXBqIf%2FIj4beU%3D	$9.660	1	$9.66		YES	YES	YES
m-20948.html	$33.07	1	$33.07	Minimum QTY Purchase 2+	YES	YES	YES
enSense/EV_ICM-20602?qs=u4fy%2FsgLU9PDvlLfBx4paA%3D%3D	$58.64	1	$58.64	Need to Extract the ICM-20602 (V15), replacement for ICM-20608-G (V14)	YES	Partial	YES
il/tdk-invensense/ICM-20608-G/5177387?utm_adgroup=General&utm_source=google&utm_medium=cpc&utm_campaign=PMax%20Shopping_Product_Zombie%20SKUS&utm_term=&productid=5177387 v-c_ext-_prd-5177387_sig-EAIaIQobChMInuXgnPCr-QIV0oJLBR2cawKjEAQYAiABEgIJfPD_BwE&gclid=EAIaIQobChMInuXgnPCr-QIV0oJLBR2cawKjEAQYAiABEgIJfPD_BwE	$11.65	1	$11.65	^In replacement for ICM-20602, this one uses PixRacer V14	YES	YES	YES
ement-Specialties/MS561101BA03-50?qs=%252BgKeJhng5iV%252BnCz6Qd5iDw%3D%3D	$16.58	1	$16.58		YES	YES	YES
NCP163ASN180T1G?qs=vLWxofP3U2z0o2tJM%252BNAYw%3D%3D	$0.76	1	$0.76		YES	YES	YES
iconductors/TJA1051TK-3118?qs=fhAOlxDPaYVrDQ2e4g%252BBJA%3D%3D	$2.10	1	$2.10	Was Expected 29/08/2022, Now 03/10/22	BACKORDER	NO	YES
8/can-transceiver-aec-q100-5mbps/dp/2776121	$3.23	1	$3.23	^^	YES	YES	YES
ncorporated/74LVC1G86SE-7?qs=8Dv1RiiPiVJ%252BnYSWg3kTgg%3D%3D	$0.53	1	$0.53		YES	YES	YES
468827	$1.12	1	$1.12	Back oder despatch 21/09/2022	YES	YES	YES
7346-0001?qs=c2CV6XM0DweJBWaSeyWeCw%3D%3D	$1.61	1	$1.61		YES	YES	YES
521800	$1.43	4	$5.71		YES	YES	YES
i-Electronics/PJS008-2120-0?qs=pDnM3xHIg41mTHT1wk7bvQ%3D%3D	$3.90	1	$3.90		YES	YES	YES
521797p?cm_mmc=AU-PPC-DS3A-_-google-_-3_AU_EN_Connectors_PCB+Headers_JST_Phrase-_-JST+-+7521797P+-+SM04B-GHS-TB(LF)(SN)-_-sm04b+ghs+tb+lf+sn&matchtype=p&aud-828197004210:kwd-349737151659 Q1AEAAYAiAAEgIOBfD_BwE&gclsrc=aw.ds	$1.17	2	$2.35		YES	YES	YES
521807p?cm_mmc=AU-PPC-DS3A-_-google-_-3_AU_EN_Connectors_PCB+Headers_JST_Phrase-_-JST+-+7521807P+-+SM05B-GHS-TB(LF)(SN)-_-sm05b+ghs+tb+lf+sn&matchtype=p&aud-821594433763:kwd-383463479017 w0kEAAYAyAAEgJjvvD_BwE&gclsrc=aw.ds	$1.21	2	$2.41	Back order despatch 30/11/2022	BACKORDER	NO	YES
SSM-105-L-DV?qs=92ilVni64gwpNieSff7mAQ%3D%3D	$5.13	1	$5.13		YES	YES	YES
ht/APG1608ZGCK?qs=Oo40KB9lBTmmOma0ledTcg%3D%3D	$0.92	1	$0.92	Green, 603 (in)	YES	YES	YES
LTST-C19HE1WT?qs=WAWaI72GHKxJLPArJA9%252BUw%3D%3D	$0.75	1	$0.75		YES	YES	YES
emiconductor/XP151A12A2MR-G?qs=AsjdqWjXhJ%252BYt8dSaKx3TA%3D%3D	$1.03	1	$1.03		YES	YES	YES
-Technologies/BSS209PW-H6327?qs=l6ZoeTYLMwPiE0APXKbLcQ%3D%3D	$0.60	1	$0.60		YES	YES	YES
CMP0603-FX-1002ELF?qs=sGAEpiMZZMtlubZbdhIBIIPWN1VQSu3wvtn3yYqkWIE%3D	$0.26	9	$2.38	10kOHM, 603 (in)	YES	YES	YES
CMP0603AFX-1004ELF?qs=sGAEpiMZZMtlubZbdhIBIIPWN1VQSu3w%252BK3x48irFUs%3D	$0.41	2	$0.82	1MOHM, 603 (in)	YES	YES	YES
ic/ERJ-UP3D1003V?qs=sGAEpiMZZMtlubZbdhIBINZyO39%252BOXIfhATqBknuev8%3D	$0.48	5	$2.42	100kOHM, 603 (in)	YES	YES	YES
Semiconductor/SDR03EZPD1200?qs=sGAEpiMZZMtlubZbdhIBIK8AeiPKd8jI5FiBmScclfw%3D	$0.32	1	$0.32	120OHM, 603 (in)	YES	YES	YES
Dale/CRCW0603220RJNEAC?qs=sGAEpiMZZMtlubZbdhIBIIZe04wfiaJWuP1eVTGiDpg%3D	$0.15	19	$2.85	220OHM, 603 (in)	YES	YES	YES
ic/ERA-3VEB1501V?qs=sGAEpiMZZMtlubZbdhIBIFX24rZb7BVEH%252BWfeSYccGI%3D	$0.97	3	$2.90	1.5kOHM, 603 (in)	YES	YES	YES
Dale/RCC06031K00FKEA?qs=sGAEpiMZZMtlubZbdhIBINZyO39%252BOXIfPVP5mE%2FLgUo%3D	$0.21	5	$1.03	1kOHM, 603 (in)	YES	YES	YES
Semiconductor/SDR03EZPD68R0?qs=sGAEpiMZZMtlubZbdhIBIK8AeiPKd8jI3Tw4pRHIIWs%3D	$0.32	1	$0.32	68OHM, 603 (in)	YES	YES	YES
CHP0603AFX-4700ELF?qs=sGAEpiMZZMtlubZbdhIBINZyO39%252BOXIfsmxjJ95ijuY%3D	$0.78	1	$0.78	470OHM, 603 (in)	YES	YES	YES
ic/ERJ-UP3F10R0V?qs=sGAEpiMZZMtlubZbdhIBINZyO39%252BOXIfP8pGmM67b0U%3D	$0.38	1	$0.38	10OHM, 603 (in)	YES	YES	YES
Semiconductor/SDR03EZPF1000?qs=sGAEpiMZZMtlubZbdhIBIIR71ikvQRXLtdTwu3hnA7E%3D	$0.21	3	$0.62	100OHM, 603 (in)	YES	YES	YES
Semiconductor/SDR03EZPD22R0?qs=sGAEpiMZZMtlubZbdhIBIK8AeiPKd8jIDpD%2FOVUe8pI%3D	$0.32	2	$0.65	22OHM, 603 (in)	YES	YES	YES
struments/LP5907QMFX-3.3Q1?qs=asCBFxFfL1RsL2VYqGFsBQ%3D%3D	$1.03	2	$2.06		YES	YES	YES
electronics/LIS3MDLTR?qs=hnsWAVXnr6EbLE48dGN9Pg%3D%3D	$5.92	1	$5.92		YES	YES	YES
iming/FA-20H-24.0000MF20X-K3?qs=CANAagUtPAp2w31393daZQ%3D%3D	$1.77	1	$1.77		YES	YES	YES
		Total	$220.16				







Info Source: https://docs.px4.io/main/en/flight_controller/pixracer
Base 2015 Schematics Source: GitHub - AUAV-OpenSource/FMUv4-PixRacer: Pixracer FlyZight Controller
Modified Base also found in Folder!

Both approaches if done correctly should be able to run the PX4 open source Firmware and connect to Mission planner or QGroundControl Software.

Key requirements: the Autopilot will be used by students, so a battery connection, PWM control, Telemetry(2x), Companioning computer connection(Serial, eth or wireless connection). SD card for logging is also required.
-Josh

Progress Reports Organised By Joshua Hecke

KAI(Wesley)----------11/10/2022

Schematics

As mentioned the version is based on the Pixhawk FMUv4 - PixRacer design.
The Schematics can be found under: **T:\Projects\QUT_UAV_Autopilot\FMUv4_Pixracer\PIXRACER-V1-R15.pdf**

PIXRACER-V1-R15

The QUT UAV Autopilot version one is designed using Altium, and all the schematics and PCB document files can be found in the folder: **T:\Projects\QUT_UAV_Autopilot\FMUv4_Pixracer\QUT_Autopilot_Pixracer**

Problems Encountered / Be Aware

- The schematic of FMUv4 – PixRacer have components that differ to the ones used in the QUT-Autopilot due to discontinued components or out of stock.
- The QUT-Autopilot IMU schematic uses ICM-20602, where the components acquired is ICM-20608-G. The pins have double checked to be the same, with very similar characteristics. So should not be a problem. If in doubt, please double check.
- Also do aware due to chip shortage, many chips are hard to get, especially the main processor chip, STM32F427VIT6. The one acquired was from all-express, unsure of its authenticity / useability. This chip was not tested yet, due to the issue with sensors.
- Aware testings are made using Arduino, some libraries are grabbed from repositories as well as Arduino library manager.
  - Most libraries should be included in file path "T:\Techs\Will_Wesley\QUT_UAV_Autopilot\FMUv4_Pixracer\Software\libraries"
    - Transfer this file to "Document\Arduino\libraries" and it should work
  - For library from Manager
    - **Sparkfun 9DoF IMU Breakout (For IMC-20948)**
- **The Magnetometer sensor LIS3MDxx has the MISO and MOSI signals placed wrong, so the board cannot be applicable for full build (DO create a new board). Have not changed the signals around in Altium, just to be consistent with the error.**

Future Improvements / Suggestions

- Preferably re-design the board layout with larger spacings, larger board and several testing points to test the sensor's signal as well as various other signals. This can help verify that the board connections are correct. (Through holes for jumper wires are recommended at least for the sensors, since it can be easily soldered and plug right into Arduino for testing).
- Maybe can design a temporary chip testing PCB to test sensors / chip integrity, and verify they are not broken.
- Maybe create a better Arduino software for testing SPI to sensors. Right now, using pre-made libraries to test, since limited understanding to SPI.

SETH----------1/6/2023

Recommendation Before Proceeding

- Read the handover documentation
- Understand the schematic
- Familiarise with the board layout
- Do recommend redesigning the board with several test points
  - Problem with current design, ever thing is small and minimal, which became an issue as first prototype

More Recommendations (Seth - 7/06/2023)

- Test the sensors that have arrived on this day, I unfortunately did not have time to test these.
- Continue to recreate the autopilot that was lost at Virginia

AnLe----------3/1/2024

Summary:
The project is forming its own path, and awaits the next "adventerer" to roam. From the handover documents, there were attempts to build the controller. However, those lack consistency, reliability, while cost were reasonably high. This run focus on researches. A promising prior project was found and approved by Steven. The components had yet to be ordered, more research will be needed on this matter. Among the purchases, ICM-20948 and a few usefull others were detected, these will potentially be be used when designing/building the controller.

Hand over saved, no real documented progress-Josh

CORY----------24/9/2024

Notes
Have not yet tested the Main processing chip STM32F427VIT6, so will also verify the integrity of the chip.
Arduino UNO powers the SPI bus with 5v while most of our IC's take 1.8 - 3.3v on their SPI pin, likely frying the sensor. Need to order a 5v to 3.3 and 1.8v level switcher
SCLK voltage is usually low as Arduino pin 13 is broken, only outputs 0.2v. With a pullup resistor of 5 ohms from power to pin 13, it should work again.

Sensor Testing

- **STM32F - Microcontroller** - Arduino 3.3V to Vin. Arduino GND to GND. Arduino Rx to Rx. Arduino Tx to Tx.
  - Input voltage 1.7 - 3.6V
- **ICM-20948 - IMU** - Arduino 3.3V to Vin. Arduino GND to GND. Arduino SCL to MPU9250 SCL. Arduino SDA to MPU9250 MOSI.
  - Input Voltage 1.7 - 3.6V, I/O pins requiring 1.71 - 1.95.
- **ICM-20608-G - IMU** - Arduino 3.3V to Vin. Arduino GND to GND. Arduino SCL to MPU9250 SCL. Arduino SDA to MPU9250 MOSI.
  - Both VDD/ VDDIO 1.71 - 3.45V
- **MS5611 - Barometer** - Arduino 5V to 5V in, Arduino 3.3v to 3.3V en, Arduino GND to GND. Arduino pin 13 to SCK, Arduino pin 12 to MISO, Arduino pin 11 to MOSI, Arduino pin 10 to CS.
  - VDD 1.8 - 3.6v
- **LIS3MD - Magnetometer** - Arduino 5V to 5V in, Arduino 3.3v to 3.3V en, Arduino GND to GND. Arduino pin 13 to SCK, Arduino pin 12 to MISO, Arduino pin 11 to MOSI, Arduino pin 10 to CS.
  - VDD 1.9 - 3.6v
- **ESP8266 - Wi-Fi Module** - Arduino pin 1 to ESP TX, Arduino pin 0 to ESP RX, Arduino 3.3v to ESP 8266_PD, Arduino 3.3v to ESP VCC, Arduino GND to ESP GND

Key Notes

- Arduino Uno powers the SPI bus with 5v while most of our IC's SPI pins aren't 5v tolerant causing issues that most likely cook the IC. Will likely need a bidirectional level shifter to step down voltages from each SPI pin or can use a voltage divider with resistors for testing purposes.
- SCLK voltage is usually low as Arduino pin 13 is broken, only outputs 0.2v. With a pullup resistor of 5 ohms from power to pin 13, it should work again.
- The old PCB had the MISO and MOSI signals placed wrong around the magnetometer. This has been changed on the Altium schematic; however, the new PCB hasn't been ordered yet.
- The existing schematic uses MPU-9250 however the used part is the ICM-20948
- Processor – STM32F427VIT6 bought from Ali-express and hasn't been tested yet.

Project directory: T:\Projects\QUT_UAV_Autopilot
Software: Arduino (For testing signals)
Hardware: Tote Boxes (QUT UAV Autopilot)
(For more details head to the project directory and locate the handover documentation)

Summary:

- First version of the board have been designed and manufactured.
- Second version is made and ready to be ordered.
- Further actions are required, please refer to the handover documentation

Joshua----------2/12/2025

2/12/2025-

Had a look at everyone's handovers and tried to understand the approach. Also had a look at the FMUv3 open source designs Aswell as there is a lot more content online with their designs. It seems that Cory had updated the original design Wesley had to a working design ready in Altium which was taken from the Poracer FMUv4 design. All the parts for it already at QUT in a box with multiple attempts at soldering, no clue when these attempts were made and what success they had. From my understanding it seems he was soldering up to half then ran into a flipped lines so he designed the fix. Spend time going over and organising the collection of parts and relating them to the Altium Design. It also seems he did a lot of testing of IC's however the his hand over is very vague on the results of the testing. Following the advice of the Last Handover document (CORY)

-

To Do:

Order new PCB from Altium schematic. (NEED TO ORDER, GET stencils ASWELL AS SMD components USED)

Buy remaining components missing. (FOUND TO BE DONE ALREADY)

Potentially make or buy breakout boards to easier test certain sensors within the circuit.

Solder up and test major sensors on new PCB.

Load Ardupilot firmware onto flight controller and begin customising.

-

Will also review the Altium design to ensure all the parts that are on there are on the list. Completed a stocktake of the parts it seems all required parts are here, However the amounts of parts is unknown need to cross reference the design to ensure there is enough parts for multi component numbers. Added Component List to OneNote so it does not get lost as it keeps track of all components in the box and is accurate as of 2/12/2025.

Need to double check that the new design has MISO MOSI lines flipped for magnetometer.
Need to ensure main chip works as no one has done that yet and there are concerns due to where it was ordered from (AliExpress). Research testing setups or approaches so everyone can stop worrying about it.
Need to ensure we have all the parts on hand before sending new PCB Order in and that the parts we have on hand match the ones in the table above(ignore colours in storage column cannot remove them)

Missing from table is the ESP wifi chip and the header for it, we do have these they are in the box along with some additional level shifters for testing with Arduino. As of right now I have no idea how the schematic works...

8/12/2025- had to update computer to win 11 so took out half the day, working on pi 5 camera python app while computer was down. After reading handover notes it says that the new board is ready to fabricate and should work.

However I spent the day investigating the magnetometer fix and the addition of the different imu(ICM-20498). The magnetometer fix should be good however will double check again later, What has me worried is the new imu has a logic level of 1.8 volts however it has direct connection to miso mosi sck int and cs which are all connected to the MCU(STM32 chip). The STM32 chip is run at 3.3 volts so all the communications are at double the expected voltage for a 1 reading(1.8) which means the chip wont work or get fried. The fix to this is adding a level converter for this new chip so all the comms lines are shifted down to 1.8 volts. To do this a new part needs to be ordered and the design needs to make room for this shifter... not easy.

I have started to convert the design to kicad so
its easier to work with and future students can work on an open source/ no paywalled software and the possibility to work on it at home opens up. During conversion some components have gone missing (3d model only) so I want to make sure everything converted over first before looking at adding shifter. THIS DID NOT WORK KICAD CONVERSION IS BUNS

Shifter Approach

Pin Name	Side A (1.8V Side)	Connection	Side B (3.3V Side)	Connection
VCCA	Input	VDD_1V8_SENSORS	VCCB	VDD_3V3_SENSORS
GND	GND	GND	GND	GND
A1	SCK	IC4 Pin 23 (SCL/SCLK)	B1	MCU Net ICM20948_SCK
A2	MOSI	IC4 Pin 24 (SDA/SDI)	B2	MCU Net ICM20948_MOSI
A3	MISO	IC4 Pin 9 (SDO/AD0)	B3	MCU Net ICM20948_MISO
A4	CS	IC4 Pin 22 (nCS)	B4	MCU Net ICM20948_CS
A5	INT	IC4 Pin 12 (INT1)	B5	MCU Net ICM20948_DRDY
OE	Enable	VDD_1V8_SENSORS (High)		

Shifter Pinout

4 Pin Configuration and Functions

Figure 4-1. RSV Package, 16-Pin UQFN (Top View)

Figure 4-2. BQB Package, 16-Pin WQFN With Exposed Thermal Pad (Top View)

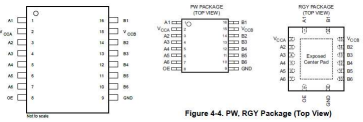
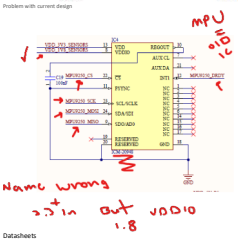


Figure 4-4. PW, RGY Package (Top View)

Problem with current design



Datasheets

DS-000189-ICM-2094... 1.8v IMU^

txb0106

Shifter^

stm32f427vg MCU^

Will ask tech to review this find to make sure I don't ruin a perfectly good design with random fixes(pcb tech not in today he is sick)

Additional Validation

The datasheet for the STM32F427VG does not explicitly state "Pin PA4 outputs 3.3V" in a single line. Instead, it defines the output voltage (V_{OH}) relative to the supply voltage (V_{DD}).
If you are powering the MCU with 3.3V (which is standard for Poracer), the **Electrical Characteristics** section confirms the outputs will drive ~3.3V.
Here is the exact location in the datasheet that defines this:

1. The Output Voltage Specification

Location: Section **6.3.17 (I/O port characteristics)**, specifically **Table 57: Output voltage characteristics**.

- **Parameter:** V_{OH} (Output high-level voltage).
- **Condition:** For a standard CMOS output with $I_{IO} = +8 mA$ (current load).
- **Specification:** The table states that $V_{OH} \ge V_{DD} - 0.4 V$.

Meaning:
If your board powers the STM32 MCU at 3.3V ($V_{DD} = 3.3V$), the Output High voltage will be at least 2.9V ($3.3V - 0.4V$). In practice, with light loads (like communicating with a sensor), it will be extremely close to the full 3.3V.

2. Validating the Pins (PA4-7, PD15, PC2)

You can confirm these pins follow standard I/O voltage rules in the **Alternate function mapping** section (typically **Table 12** around page 75 in standard revisions).

- **PA4 - PA7:** These are configured as **SPI1** (Alternate Function 5).
  - **PA4:** SPI1_NSS (Chip Select)
  - **PA5:** SPI1_SCK (Clock)
  - **PA6:** SPI1_MISO (Master In Slave Out)
  - **PA7:** SPI1_MOSI (Master Out Slave In)
- **PD15 & PC2:** These are being used as standard **GPIOs** (likely for Chip Select or Data Ready).

Since the STM32 is powered by 3.3V, all these pins (SCK, MOSI, CS) will output 3.3V signals. This confirms the conflict: You cannot connect these 3.3V lines directly to the ICM-20948 if its VDDIO is tied to 1.8V, because the sensor is not 3.3V tolerant on digital lines when configured for 1.8V operation.
From <gemini 3.0>

9/12/2025- have confirmed that the current board design will fry the ICM-20948 with techs and the current design is obsolete. Looking at next two approaches

1-start again from scratch and ensure replacements are 3.3 tolerant for compatibility
2-Add level shifters for io pins using above device

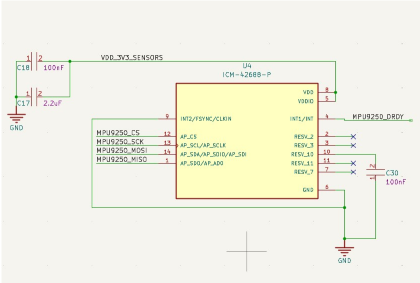
Looking at a approach which replaces current sensor with 3.3v tolerant one. Got the symbol wired up but now having trouble with wiring footprint on software(kicad import super messy and Altium wont let me edit :(.(FIXED HAD TO ACTIVATE LICENCE)

ICM-42688-P

ICM_42688_P_Altium

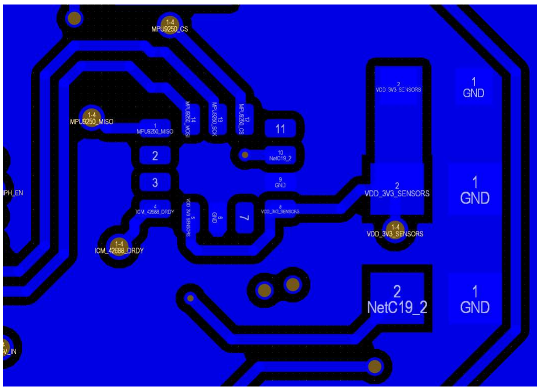
ICM_42688_P

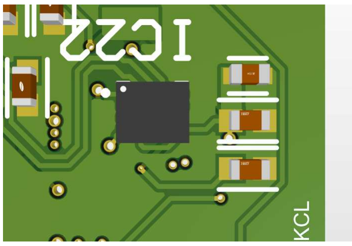
Below is the symbol and wiring for the new part in kicad, this was more just to confirm new wiring and kicad is too messy to be used an export target. Moving into learning how to
A-duplicate the original Altium files for security
B-Import new symbol
C-delete old IC7 U4 from schematic and pcb layout
D-add new wiring diagram
E-Update PCB LAYOUT
F-review whole board again



^kicad practice schematic

After Altium lessons from yt I have now removed the IC4 and IC7 and replaced with the new IC called IC22





15/12/2025-

TASKS

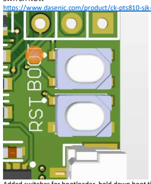
1- Reporting on onenote
2- Finish imu redesign DONE NO WORK NEEDED
3- add button for code flashing (need to pick button) DONE
4-annotate board better for better communication
5- review whole design
6- create gerber files for design
7-create firmware for new board(INSTRUCTIONS READY)
8-count part numbers to ensure we have stock (NOT DONE)
9-order new parts(2 New Parts Ordered, Stock Amount not done)

MAYBE
- kicad up a tester board for main ic so we can have easier debugging/ code execution(after everything is done)

----

need to ensure ground on new ic is connected first (DONE)
https://www.dasenic.com/product/tdk-invensense-icm-42688-p-8880834?srsltid=AfmBOopk_SbRC0Yi9AN4e_rENBlJrckwhcawvzn9K-9sEanbEVi62sx53gI
DONE

SWITCH NOW
https://www.dasenic.com/product/ck-pts810-sjk-250-smtr-lfs-7315346



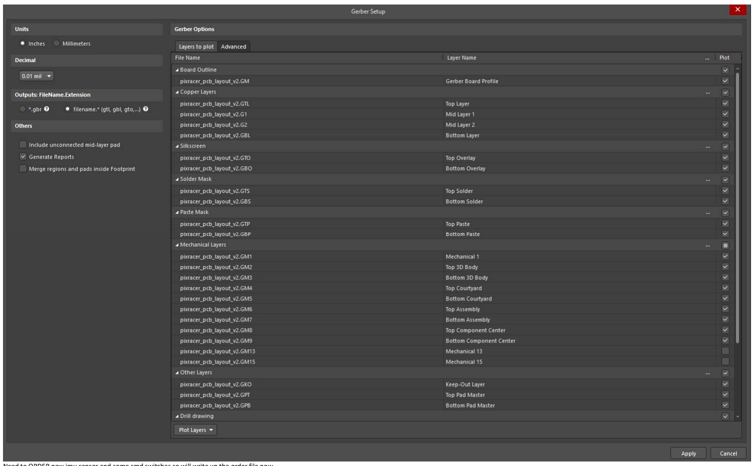
Added switches for bootloader, hold down boot then click reset to flash new code, would have been very difficult/ annoying without

ANNOTATIONS

DONE

REVIEW
DRC has a bunch of placement errors but these can be relaxed or ignored, consider looking into if we get sensor noise in the future

Gerber



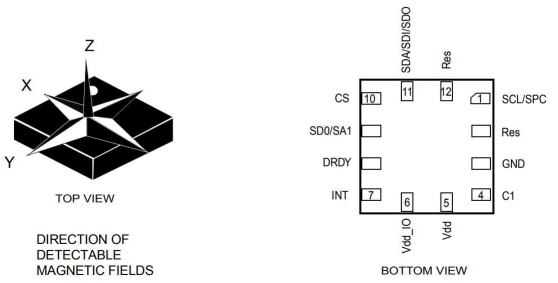
Need to ORDER new imu sensor and some smd switches so will write up the order file now.
Also will get the board firmware ready with modifications to compilation to ensure new sensors are read properly. Also need to ensure we have enough physical parts for design

Updated BOM has been made can be found in v2 output files. Need to place order for IC22 and S1,2 (BOOT RST)

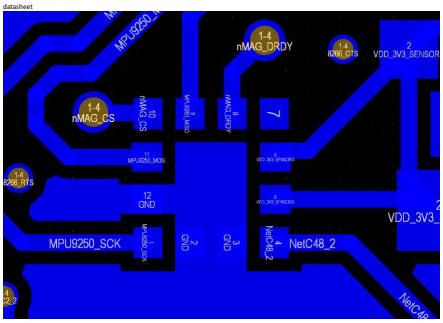
16/12/2025-

1-onenote document
2.1-review design and consider changes that will make testing easier(test points, new pcb for testing?)
2.1.1- WHAT THE HELL Is going on with the LIS3MDLTR sensor it has been flagged by 2 people fixed by 1 and still not correct??

Figure 2. Pin description

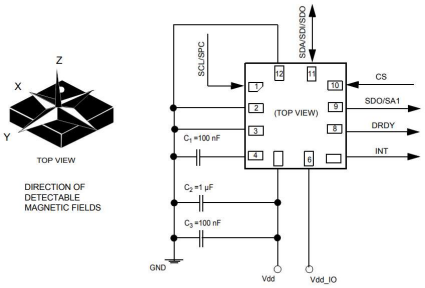


datasheet



current config?? Why inverted?? Need to look into it, wiring seems correct but the part physically won't work with this footprint

Figure 5. LIS3MDL electrical connections



This makes more sense, they went off application config
First image was bottom view, problem solved!

2.2-Order PCB and parts

order put through for ic icm-42688 and switches along with v2 board. Fingers crossed the board has not many issues(it did)

JOSHUA_HECKE_WIL

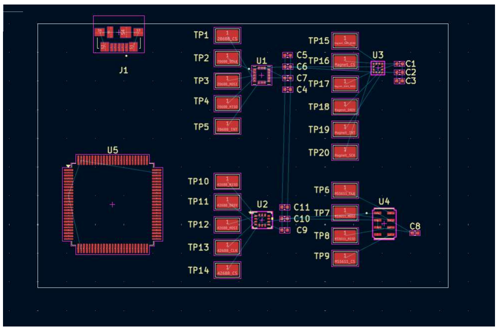
T5 EE&I Order Template v3

Name	Joshua Hecke	**Unit**	QUT AutoPilot Research	**Do not leave empty cells in the descriptions.**	
Email	josh.hecke@qut.edu.au	**Group**	WIL	**Do not change formatting or file type.**	
Account	324530/4111 (EE&I Consumables)			**Complete this form and email to foe.eei@qut.edu.au**	
Supervisor	Brendon Stichbury	**Multiplier**	1	**Also cc your supervisor who will authorise the order**	

Item	Description	Quantity	Cost Each	Currency	Supplier	Order Code	Web Link
1	TDK InvenSense ICM-42688-P	5	$4.30	USD?	Dasenic	I688-P-DS	https://www.dasenic.com/product/tdk-invensense-icm-42688-p-8880834
2	C&K PTS810 SJK 250 SMTR LFS	6	$0.31	USD?	Dasenic	MTR LFS-DS	https://www.dasenic.com/product/ck-pts810-sjk-250-smtr-lfs-7315346
3	PCBWAY Order WS4332AS53401 + Stencil	5		USD?	PCBWAY	332AS53401	N/A

Order issues so fixing them up now (PCBWAY email rep, im thinking its drill hole settings that were not communicated properly)

Mean time working on designing a test board for the sensors and the MCU so the smd sensors can be tested solo with the bare min parts then integrated all together later. This is a mini project with the main goal is to improve circuit design and datasheet reading skills while main design is being fabricated. Key hardships are foot print selection and considering key choices like how am I flashing code to this board, how am I powering it? How to wire up a MCU like the STM?
Done in Kicad



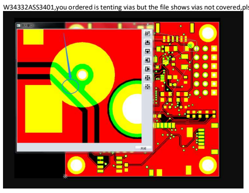
need to add MCU connection and button to finish design. Consider what parts are needed for usb power to board power connection. Goal is to test spi comms with all sensors, then use success to build main board

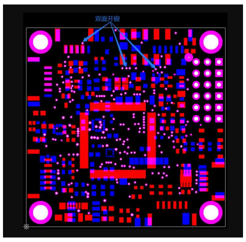
22/12/2025-

PCB failed during review before manufacturing :(,here is output from PCB techs.

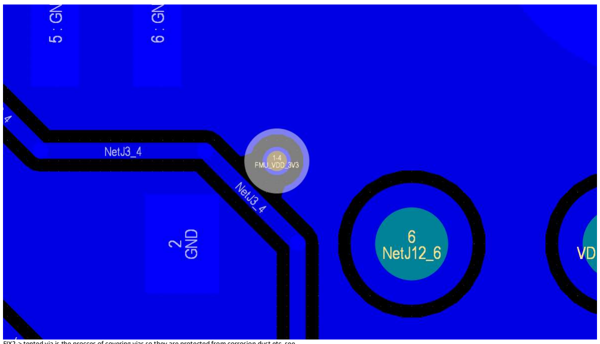
Feedback from PCBway:

W34332ASS3401,you ordered is tenting vias but the file shows vias not covered,pls advise if we review follow the file. The file includes exposed traces,pls advise if the file is correct





FIX1-> fix exposed tracer

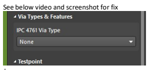


FIX2-> tented via is the procces of covering vias so they are protected from corrosion,dust etc. see

he idea behind via tenting is simple: you're covering any vias in your PCB with solder mask so that any pad/ring on the via hole, and the via barrel itself, are not exposed to the environment. The solder resist is placed to provide some measure of protection for the via pad and the plating inside the via barrel. If you look at a PCB layout, you can spot tented vias just by looking at your solder stop mask layer; the same applies to the Gerber file for that solder mask layer.

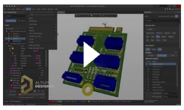
From <https://resources.altium.com/p/when-use-tented-vias-your-pcb-layout>

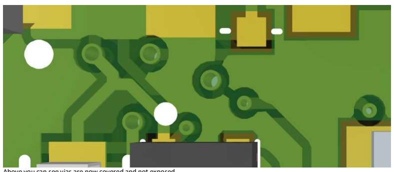
See below video and screenshot for fix



+

https://www.youtube.com/watch?v=dJ3TLudgPwg





Above you can see vias are now covered and not exposed

The new files contains the fixes required, sent off for evaluation by manufacture, more progress on KICAD tester board done, added buttons and a required regulator to get the needed 3.3 volts. Need to now wire STM correct and connect SPI lines. It's a 4 layer board so this is practice with routing techniques ensure its tented.

New pcb has been ordered

23/12/2025-

Issues with pcb manufacturing again, below is the output

Hi,
About the pcb order W34332ASS3404,there are 3 questions need to confirm:
1.For the board thickness to hole diameter ratio is greater than 10:1, it is recommended to enlarge the hole size,pls revise the file.

2. Pls confirm if you need to do the soldermask bridge?
If not,we will do soldermask opening for the whole IC parts,ok?
If yes,we require the IC pin spacing needs to be 0.19mm with green soldermask ink,pls change it.

3. The spacing between different net about hole size should be larger than 0.3mm,pls revise the file then send the new file for check.

Really stuck on these fixes as I really don't know that they mean with their 3 points?

It sounds like your manufacturer's "Engineering Queries" (EQ) are pointing out some technical limitations of their production line compared to your current design.
Here is a breakdown of what these points mean and how you should address them in your PCB design software (like Altium, KiCad, or Eagle).

1. Aspect Ratio (Thickness to Hole Diameter)

The Issue: The manufacturer is concerned about the **Aspect Ratio**. If a board is 1.6mm thick and your drill hole is only 0.1mm, the ratio is 16:1. It is very difficult for the plating chemicals to flow through such a tiny, deep "tunnel" to coat the inside with copper.

- **What to do:** * Check your smallest via/hole size. If your board is **1.6mm thick**, try to ensure no hole is smaller than **0.16mm** (to keep it at 10:1).
  - **Action:** Either increase the diameter of your smallest vias or ask the manufacturer if they can handle a higher aspect ratio for an extra fee.

2. Soldermask Bridges vs. Openings

The Issue: A "soldermask bridge" is the thin strip of green mask between two SMD pads (like the pins on an IC). This bridge prevents solder from flowing between pins and causing a short circuit (bridging).

- **The Problem:** Their machines can't print a "bridge" thinner than about 0.1mm. If your IC pins are very close together, there isn't enough room for that green strip.
- **The Options:**
  - **Option A (Gang Mask):** You remove the bridges. The entire IC area becomes one big "opening." This is riskier for soldering because bridges won't be there to stop shorts.
  - **Option B (Modify Design):** You change your footprint so the space between the copper pads is at least **0.19mm**. This gives them enough room to print a reliable solder mask bridge.
- **Action:** I recommend **Option B**. Check your IC footprints and ensure the gap between pads is at least 0.19mm.

3. Hole-to-Hole Spacing (Different Nets)

The Issue: When two holes (vias or leaded components) belonging to different electrical signals (nets) are too close together, the drill bit might "drift" or the copper plating might bleed, causing a short circuit.

- **What to do:** * The manufacturer wants at least **0.3mm of clearance** between the edges of any two holes that aren't connected to the same signal.
  - **Action:** Run a **Design Rule Check (DRC)** in your software with the "Hole to Hole" clearance set to 0.3mm. Move any vias or pads that are flagging as too close.

Summary of what you need to send back:

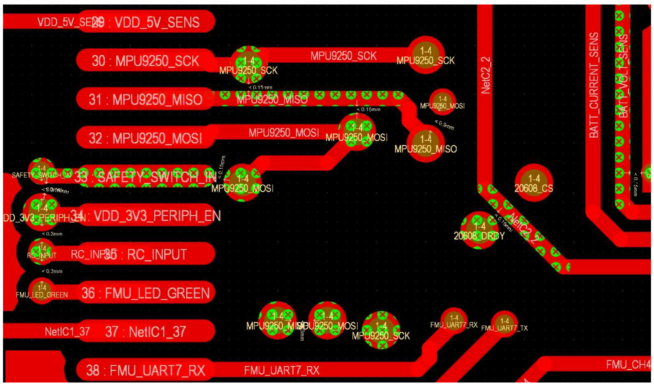
The manufacturer has asked for a **revised file**. You cannot just tell them "okay"; you need to:

1. Adjust your via sizes (Point 1).
2. Adjust IC pad spacing (Point 2).
3. Increase clearance between holes (Point 3).
4. Export new **Gerber files** and send them the updated version.

Would you like me to explain how to set these specific design rules in a particular software like Altium, KiCad, or Eagle?

From <https://gemini.google.com/app?is_sa=1&is_sa=1&android-min-version=301356232&ios-min-version=322.0&campaign_id=bkws&utm_source=sem&utm_source=google&utm_medium=paid-media&utm_medium=cpc&utm_campaign=bkws&utm_campaign=2024enAU_gemfeb&pt=9008&mt=8&ct=p-growth-sem-bkws&gclsrc=aw.ds&gad_source=1&gad_campaignid=22437964261&gbraid=0AAAAApk5BhmU7PNLqCKu3Oaet8PTbyiFH&gclid=EAIaIQobChMI4uOh1L2dkQMVVBaDAx2PWRy5EAAYASAAEgIOofD_BwE>

I have modified the rules to fit the manufactures needs and run a design rule check to help me identify what parts are failing



There are around 370 failures so need to start fixing

ISSUE1->HOLE SIZE
The board is 1.6mm thick so all holes need to be 1.6m or greater, made this a design rule to find the 14 that were 1.27 wide and fixed them. 10
FIXED

ISSUE2->Soldermask spacing
This confuses me, as the advice says 0.19 space but our IC1 has pins 0.1 spaced apart so how is this possible?
NOT FIXED

ISSUE3-> holes cannot be 0.3 close to each other, this was made a rule and DRC ed to find all holes that were too close.
FIXED

Possible issue4-> silkscreen covering pads?

05/01/2026-

New year same project. All WIL students have lost all access so there will be a period of time where work is slowed down while access is restored.(visitor accounts were removed)
A-fix the 0.19 spacing constraint and get board shipped off
B-Wire the kicad tester board and get it shipped off aswell

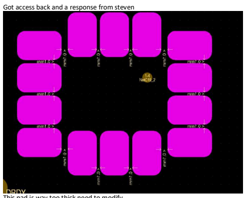
notes-> 2 boards have failed to pass inspection due to issues, I don't know if both of those board were paid for, then error occurs so payment was made but the order is in limbo or payment was not made so order can be restarted.
IF B need to email the original order back with the new files(Brendon needs to email as I have even less access now).
IF A create new order with above fixes and potentially add the tester board into the order Aswell!

thinking of waiting till the 5 day meeting to communicate this as he is very busy over the next couple days and these issues seem out of scope. Still need to ask steven about ISSUE2 as I don't know how to fix this constraint.

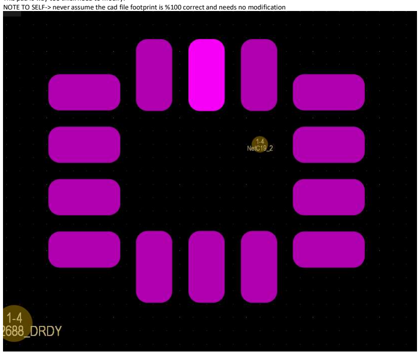
Steven came and reviewed the issues, said this was weird and could be due to how it was inputted in pcb way. He said send it too him via email and he will see how its supposed to inputted into pcbway (thank you steven you're the best). While we wait moving onto improving and wiring kicad tester board which is my design.

06/01/2026-

Got access back and a response from steven



This pad is way too thick need to modify.
NOTE TO SELF-> never assume the cad file footprint is %100 correct and needs no modification



New pads improved
Design file below

QUT_Autop
ilot_Pixra...

Waiting for Steve to contact pcbway with a update

This is a great start for a sensor breakout and MCU test platform. The **STM32F427VIT6** is a high-performance chip, but it has very specific power requirements that must be met for it to boot and run reliably.
Below is a detailed review and a step-by-step guide

A. Design Review & Suggested Changes

1. Critical: VCAP Capacitors

The STM32F4 series has an internal voltage regulator that powers the CPU core. You **must** add external capacitors to the VCAP pins for the MCU to function.

- **Action:** Connect a **2.2µF** (Low ESR ceramic) capacitor from **VCAP_1** (Pin 49) to GND.
- **Action:** Connect a **2.2µF** (Low ESR ceramic) capacitor from **VCAP_2** (Pin 73) to GND.
- *Note: Do not connect these pins to 3.3V; they are output pins for the internal regulator.*

2. MCU Decoupling Capacitors

Your current schematic for **U5** does not show decoupling capacitors.

- **Requirement:** Every VDD pin (Pins 11, 19, 28, 50, 75, 100) and VDDA (Pin 22) needs a **0.1µF** capacitor placed as close to the pin as possible on the PCB.
- **Bulk Cap:** Add at least one **4.7µF** or **10µF** tantalum or ceramic capacitor to the 3.3V rail near the MCU.

3. SWD Programming Header (Missing)

While you have USB for programming via DFU mode, it is highly recommended to add an **SWD (Serial Wire Debug)** header. This allows you to use an ST-Link for real-time debugging and "unbricking" the chip.

- **Pins needed:** GND, 3.3V, SWDIO (PA13 / Pin 72), SWCLK (PA14 / Pin 76), and NRST (Pin 14).

4. USB ESD Protection

Since this board involves manual handling and plugging/unplugging, the USB data lines (D+/D-) are vulnerable to static.

- **Suggestion:** Add a **USBLC6-2SC6** or similar ESD protection diode between the USB connector and the MCU.

B. Finishing the Schematic Wiring

To finalize the board for programming and power, follow these specific wiring instructions for your switches and USB port.

1. Reset Switch (S1)

The reset pin (NRST) is active-low.

- **Step 1:** Connect **U5 Pin 14 (NRST)** to one side of switch **S1**.
- **Step 2:** Connect the other side of switch **S1** to **GND**.
- **Step 3:** Add a **0.1µF** capacitor from NRST to GND (to prevent noise resets).
- **Step 4:** Add a **10kΩ** pull-up resistor from NRST to **+3.3V**.

2. Boot Switch (S2)

This switch allows you to enter "DFU Mode" for programming over USB.

- **Step 1:** Connect **U5 Pin 94 (BOOT0)** to the center pin of a 3-pin slide switch (or a push button to 3.3V).
- **Step 2:** Connect one side of the switch to **GND** through a **10kΩ** resistor (Normal boot mode).
- **Step 3:** Connect the other side of the switch to **+3.3V** (System Memory boot mode / USB Programming).
- **Step 4 (Logic):** To program via USB, hold the BOOT0 switch to 3.3V while resetting or powering on the board.

3. Micro USB (J1) & Power (U6)

- **Power (VBUS):** Connect **J1 Pin 1 (VBUS)** to **U6 Pin 1 (IN)**. This takes the 5V from USB and regulates it to 3.3V.
- **Data Lines:** * Connect **J1 Pin 2 (D-)** to **U5 Pin 70 (PA11)**.
  - Connect **J1 Pin 3 (D+)** to **U5 Pin 71 (PA12)**.
- **ID Pin:** Leave **J1 Pin 4 (ID)** floating.
- **Ground:** Connect **J1 Pin 5 (GND)** to your common GND plane.
- **Shield:** Connect the USB housing pins to GND through a **1MΩ** resistor in parallel with a **4.7nF** capacitor (this helps with EMI/Ground loops).

C. Hardware Checklist for PCB Layout

Once the schematic is finished and you move back to the PCB:

1. **Ground Plane:** Use a solid ground pour on the bottom layer.
2. **Differential Pairs:** Keep the USB D+ and D- traces close together and of equal length.
3. **Sensor Placement:** You have done a good job spacing the sensors (U1-U4). Ensure the MS5611 (U4) is not placed directly under a heat source, as it is a sensitive barometric pressure sensor.

Above to be done list

Steve Advice
Ground plane on bottom, use vias to get there
2 layers
LEDs on the board for debugging

BOARD BEING MADE YESSSSSSSSSSSSSSSSSSSSSSSSS THANK YOU STEVE

12/01/2026-

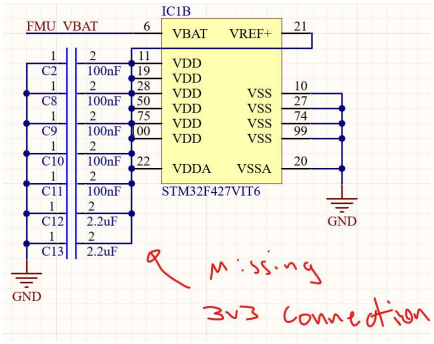
Board came but plan to start tmr when steve is ready, did kit (241) building and helped solve issues with the locker system which has been deactivated for some time now

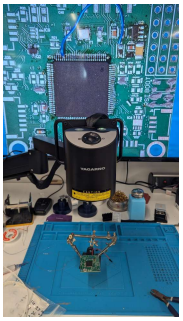
13/01/2026-

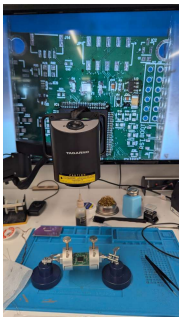
Started soldering STM and all power components, no issues so far lots of learning

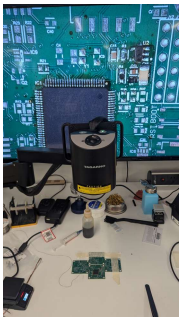
14/01/2026-

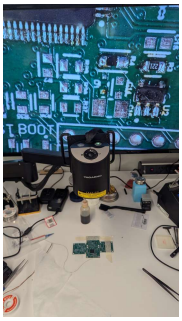
Came in today to keep powering thru on it, got led working and power in thru wires working, got stuck as usb connection not picking put when plugged it(need to debug now), major issue with design found as stm chip is missing a 3v3 input voltage (when the design was transferred from open source to this version a single line was missed which caused 3v3 to not be connected , reviewing opensource documents proved this. I should have picked it up tbf though)











19/01/2026-

Debugging the connection for usb

Possible issues I can think of
-bad physical connection on usb, stm , resistors R6 R7, Y1, C3,C4
-stm not being powered due to jumper wire failure
-data lines designed so poorly that the D+ D- out of sync(different impedience)
-resistors on data line failed
-buttons not working so board never enters DFU mode
-Stm chip from alibaba maybe its fake and we need a new one
-drivers or usb cord not correct(computer issue)

CHECKING MODE
1-Bad connections looking at switch first

SOLVED IT WAS A SHORT OVER BOOT0 PULL DOWN RESISTOR IM A GOD

```
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 003 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 004 Device 002: ID 30c9:0062 Luxvisions Innotech Limited Integrated Camera
Bus 004 Device 003: ID 8087:0033 Intel Corp. AX211 Bluetooth
Bus 003 Device 004: ID 0483:df11 STMicroelectronics STM Device in DFU Mode
Bus 004 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

DFU mode working and lets look at programming LED!!

Downloading stm32cubeprogrammer and learning about dfu mode

Downloaded stm32 MX,IDE,PRG
Used them to flash C code to the stm chip all so far so good,

Led red and green light programmed using simple pin high pin low code, cannot get blue for the life of me, Don't know if it's a hardware or software. Going to spend tmr learning how the led works so debugging steps make more sense

Good progress tho

20/01/2026-

New day refreased view, with in 10 minutes noticed I had the part 90 degreas wrong, once roated all worked well!! LED are low on so make the pin low for it to turn on and the others high for only one colour.
moving onto soldering my IMC component, this is trick as I cannot use stencil due to parts already on the board so need to hand place the paste for the small ic pins, will take a lot of time and care so no rushing, once this is tested will start a new board with full application of components on a v3 board

R5 missing from BOM investigating
-> we have it same as R3 for now, different specs tho

I am the best to ever do it! I have my selected ICM-42688-p soldered and working on the board, led led changes colour based on different tilt angles
Use STm32 MX,IDE,PRG for stm chip programming in C

going to keep populating and testing board, look at uart for testing.

RAN OUT OF ICS NEED TO REORDER
OTHER SENSORS SPI NOT WORKING(could be physical soldering issue, Programming or Other like no power due to missing compoenets)

27/01/2026-

Looking at trying to communicate with other 3 sensors on board today.

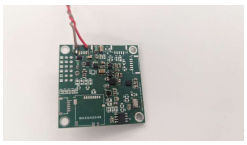
Got 2/3 working, just needed to ensure solder connections is good. Baro on the otherhand is giving me a headache. All IC6 baro sensors are probs damanged and so NEED MORE IC6. Extra extra care when soldering was needed and not done mb.

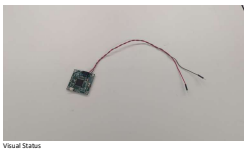
28/01/2026-

Barometer was connected to a spi2(fram_sck etc). Once I turned on spi2 it communicated, have not tested the readings(most likey broken due to hot air gun but not important right now), stuck on Fram communication right now

Got it working need a older model ID register check !

moving on to ESp8266 communication for debugging as no serial wire(we could use usb but with DFU mode taking up that its gonna be a pain plus potential to damage)





Visual Status

Below code is sensor testing code for stm chip, requires stm32MX to create the base for code(my input is in USER CODE 2 Section) and STM32IDE to compile to a Elf file which can then be uploaded into STM32 programmer app. Ill let you figure that out because If I could figure it out you can too <3.

```c
/* USER CODE BEGIN Header */
/**
  ******************************************************************************
  * @file           : main.c
  * @brief          : Main program body
  ******************************************************************************
  * @attention
  *
  * Copyright (c) 2026 STMicroelectronics.
  * All rights reserved.
  *
  * This software is licensed under terms that can be found in the LICENSE file
  * in the root directory of this software component.
  * If no LICENSE file comes with this software, it is provided AS-IS.
  *
  ******************************************************************************
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"
/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */
/* USER CODE END Includes */
/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */
/* USER CODE END PTD */
/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */
/* USER CODE END PD */
/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */
/* USER CODE END PM */
/* Private variables ---------------------------------------------------------*/
SPI_HandleTypeDef hspi1;
SPI_HandleTypeDef hspi2;
/* USER CODE BEGIN PV */
int16_t x_offset = 0;
int16_t y_offset = 0;
/* USER CODE END PV */
/* Private function prototypes -----------------------------------------------*/
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_SPI1_Init(void);
static void MX_SPI2_Init(void);
/* USER CODE BEGIN PFP */
/* USER CODE END PFP */
/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
/* USER CODE END 0 */
/**
  * @brief  The application entry point.
  * @retval int
  */
int main(void)
{
  /* USER CODE BEGIN 1 */
  /* USER CODE END 1 */
  /* MCU Configuration--------------------------------------------------------*/
  /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
  HAL_Init();
  /* USER CODE BEGIN Init */
  /* USER CODE END Init */
  /* Configure the system clock */
  SystemClock_Config();
  /* USER CODE BEGIN SysInit */
  /* USER CODE END SysInit */
```

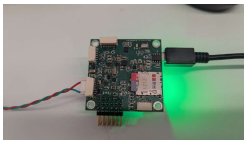
```
  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_SPI1_Init();
  MX_SPI2_Init();
  /* USER CODE BEGIN 2 */

  /* 1. SPI BUS SILENCE & POWER-UP */
  // Silence SPI1 (Main Sensors)
  HAL_GPIO_WritePin(GPIOC, ICM_CS_Pin, GPIO_PIN_SET);       // PC2 [cite: 152]
  HAL_GPIO_WritePin(GPIOC, ICM2_CS_Pin, GPIO_PIN_SET);      // PC15 [cite: 217]
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_SET);       // PE14 [cite: 284]
  // Silence SPI2 (Baro & FRAM) - CRITICAL to prevent bus noise
  HAL_GPIO_WritePin(GPIOD, BARO_CS_Pin, GPIO_PIN_SET);      // PD7 [cite: 177]
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);      // PD10 (Corrected FRAM_CS)
  // LEDs OFF (Active-Low)
  HAL_GPIO_WritePin(GPIOB, GREEN_LED_Pin|RED_LED_Pin|BLUE_LED_Pin, GPIO_PIN_SET);
  // Enable sensor power rails
  HAL_GPIO_WritePin(GPIOE, VDD_3V3_SENSORS_EN_Pin, GPIO_PIN_SET); // PE3 [cite: 233]
  HAL_GPIO_WritePin(GPIOC, VDD_3V3_PERIPH_EN_Pin, GPIO_PIN_SET);  // PC4 [cite: 165]
  HAL_Delay(200); // Stabilization time
  uint8_t id_reg, id_val;
  uint16_t C[7]; // MS5611 Calibration coefficients
  /* --- SPI1 TESTS --- */
  // ICM-42688-P Test
  uint8_t reset_42688[2] = {0x11, 0x01}; // Soft Reset
  HAL_GPIO_WritePin(GPIOC, ICM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, reset_42688, 2, 10);
  HAL_GPIO_WritePin(GPIOC, ICM_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(50);
  id_reg = 0x75 | 0x80;
  HAL_GPIO_WritePin(GPIOC, ICM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, &id_reg, 1, 10);
  HAL_SPI_Receive(&hspi1, &id_val, 1, 10);
  HAL_GPIO_WritePin(GPIOC, ICM_CS_Pin, GPIO_PIN_SET);
  if (id_val != 0x47) { while(1) { HAL_GPIO_TogglePin(GPIOB, RED_LED_Pin|GREEN_LED_Pin); HAL_Delay(200); } }
  // ICM-20608 Test
  HAL_GPIO_WritePin(GPIOC, ICM2_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, &id_reg, 1, 10);
  HAL_SPI_Receive(&hspi1, &id_val, 1, 10);
  HAL_GPIO_WritePin(GPIOC, ICM2_CS_Pin, GPIO_PIN_SET);
  if (id_val != 0xAF) { while(1) { HAL_GPIO_TogglePin(GPIOB, GREEN_LED_Pin|BLUE_LED_Pin); HAL_Delay(200); } }
  /* 4. SENSOR 3: LIS3MDL MAGNETOMETER (PE14) */
  // Step A: Soft Reset & Reboot Memory
  uint8_t mag_reset[2] = {0x21, 0x04 | 0x08}; // SOFT_RST + REBOOT [cite: 1611, 1612]
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, mag_reset, 2, 10);
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(50);
  // Step B: WAKE UP - Set Operating Mode to Continuous [CRITICAL]
  // CTRL_REG3 (22h), MD[1:0] = 00
  uint8_t mag_wakeup[2] = {0x22, 0x00};
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, mag_wakeup, 2, 10);
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(10); // Small settle time
  // Step C: Read WHO_AM_I (0Fh)
  id_reg = 0x0F | 0x80; // bit 0 = 1 for READ [cite: 1490]
  id_val = 0;
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi1, &id_reg, 1, 10);
  HAL_SPI_Receive(&hspi1, &id_val, 1, 10);
  HAL_GPIO_WritePin(GPIOE, MAG_CS_Pin, GPIO_PIN_SET);
  if (id_val != 0x3D) { // Expected value is 0x3D [cite: 1558, 1584]
      while(1) {
          HAL_GPIO_TogglePin(GPIOB, RED_LED_Pin | BLUE_LED_Pin);
          HAL_Delay(200);
      }
  }
  /* --- SPI2 TESTS (BARO & FRAM) --- */
  // MS5611 Barometer Reset & PROM Check
  /* 5. MS5611 BAROMETER (SPI2) */
  uint8_t baro_reset = 0x1E;
  HAL_GPIO_WritePin(GPIOD, BARO_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi2, &baro_reset, 1, 10);
  HAL_GPIO_WritePin(GPIOD, BARO_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(20);
  // Read PROM Calibration Coefficients C1-C6
  for (uint8_t i = 1; i <= 6; i++) {
      uint8_t cmd = 0xA0 + (i * 2);
      uint8_t data[2];
      HAL_GPIO_WritePin(GPIOD, BARO_CS_Pin, GPIO_PIN_RESET);
      HAL_SPI_Transmit(&hspi2, &cmd, 1, 10);
      HAL_SPI_Receive(&hspi2, data, 2, 10);
      HAL_GPIO_WritePin(GPIOD, BARO_CS_Pin, GPIO_PIN_SET);
      C[i] = (data[0] << 8) | data[1];
  }
  if (C[1] == 0 || C[1] == 0xFFFF) { HAL_GPIO_WritePin(GPIOB, RED_LED_Pin, GPIO_PIN_RESET); while(1); }
  /* --- FM25V02A FRAM SYSTEM TEST (PD10) --- */
  // 1. Recovery & Wakeup
  // Ensures the chip exits Sleep Mode (B9h) or a mid-transaction hang.
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_RESET);
  HAL_Delay(1);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(5); // tREC recovery time
  // 2. Identity Verification (RDID)
  // Scans for Manufacturer ID 0x04 (Cypress) or 0xC2 (Ramtron)
  uint8_t rdid_tx[12] = {0x9F};
  uint8_t rdid_rx[12] = {0};
  uint8_t fram_identified = 0;
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_TransmitReceive(&hspi2, rdid_tx, rdid_rx, 12, 50);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);
  for (int i = 1; i < 12; i++) {
      if (rdid_rx[i] == 0x04 || rdid_rx[i] == 0xC2) {
          fram_identified = 1;
          break;
      }
  }
  if (!fram_identified) {
      while(1) { // Error: Fast Red Flash (ID Mismatch)
          HAL_GPIO_TogglePin(GPIOB, GPIO_PIN_11); // RED_LED
          HAL_Delay(100);
      }
  }
  // 3. Memory Integrity Check (Write/Read)
  // Verifies that the memory cells are non-volatile and writable.
  uint8_t test_addr[2] = {0x00, 0x01}; // Address 0x0001
  uint8_t test_data = 0x8C;            // Test pattern
  uint8_t read_back = 0;
  uint8_t cmd_wren = 0x06;
  uint8_t cmd_write[4] = {0x02, test_addr[0], test_addr[1], test_data};
  uint8_t cmd_read[3] = {0x03, test_addr[0], test_addr[1]};
  // Write Sequence
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi2, &cmd_wren, 1, 10);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi2, cmd_write, 4, 10);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);
  HAL_Delay(5);
  // Read Sequence
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_RESET);
  HAL_SPI_Transmit(&hspi2, cmd_read, 3, 10);
  HAL_SPI_Receive(&hspi2, &read_back, 1, 10);
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin, GPIO_PIN_SET);
  if (read_back != test_data) {
      while(1) { // Error: Flash Blue (Write/Read Failure)
          HAL_GPIO_TogglePin(GPIOB, GPIO_PIN_0); // BLUE_LED
          HAL_Delay(200);
      }
  }

  // SUCCESS: SOLID WHITE
  HAL_GPIO_WritePin(GPIOB, RED_LED_Pin|GREEN_LED_Pin|BLUE_LED_Pin, GPIO_PIN_RESET);
  HAL_Delay(2000);
  HAL_GPIO_WritePin(GPIOB, RED_LED_Pin|GREEN_LED_Pin|BLUE_LED_Pin, GPIO_PIN_SET);
  /* USER CODE END 2 */
  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */
    /* USER CODE BEGIN 3 */
  }
  /* USER CODE END 3 */
}
/**
  * @brief System Clock Configuration
  * @retval None
  */
void SystemClock_Config(void)
{
  RCC_OscInitTypeDef RCC_OscInitStruct = {0};
  RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
  /** Configure the main internal regulator output voltage
  */
  __HAL_RCC_PWR_CLK_ENABLE();
  __HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE1);
  /** Initializes the RCC Oscillators according to the specified parameters
  * in the RCC_OscInitTypeDef structure.
  */
  RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSE;
  RCC_OscInitStruct.HSEState = RCC_HSE_ON;
  RCC_OscInitStruct.PLL.PLLState = RCC_PLL_ON;
  RCC_OscInitStruct.PLL.PLLSource = RCC_PLLSOURCE_HSE;
  RCC_OscInitStruct.PLL.PLLM = 12;
  RCC_OscInitStruct.PLL.PLLN = 180;
  RCC_OscInitStruct.PLL.PLLP = RCC_PLLP_DIV2;
  RCC_OscInitStruct.PLL.PLLQ = 4;
  if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
  {
    Error_Handler();
  }
  /** Activate the Over-Drive mode
  */
  if (HAL_PWREx_EnableOverDrive() != HAL_OK)
  {
    Error_Handler();
  }
  /** Initializes the CPU, AHB and APB buses clocks
  */
  RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                              |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
  RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_PLLCLK;
  RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
  RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV4;
  RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV2;
  if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_5) != HAL_OK)
  {
    Error_Handler();
  }
}
/**
  * @brief SPI1 Initialization Function
  * @param None
  * @retval None
  */
static void MX_SPI1_Init(void)
{
  /* USER CODE BEGIN SPI1_Init 0 */
  /* USER CODE END SPI1_Init 0 */
  /* USER CODE BEGIN SPI1_Init 1 */
  /* USER CODE END SPI1_Init 1 */
  /* SPI1 parameter configuration*/
  hspi1.Instance = SPI1;
  hspi1.Init.Mode = SPI_MODE_MASTER;
  hspi1.Init.Direction = SPI_DIRECTION_2LINES;
  hspi1.Init.DataSize = SPI_DATASIZE_8BIT;
  hspi1.Init.CLKPolarity = SPI_POLARITY_HIGH;
  hspi1.Init.CLKPhase = SPI_PHASE_2EDGE;
  hspi1.Init.NSS = SPI_NSS_SOFT;
  hspi1.Init.BaudRatePrescaler = SPI_BAUDRATEPRESCALER_16;
  hspi1.Init.FirstBit = SPI_FIRSTBIT_MSB;
  hspi1.Init.TIMode = SPI_TIMODE_DISABLE;
  hspi1.Init.CRCCalculation = SPI_CRCCALCULATION_DISABLE;
  hspi1.Init.CRCPolynomial = 10;
  if (HAL_SPI_Init(&hspi1) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN SPI1_Init 2 */
  /* USER CODE END SPI1_Init 2 */
}
/**
  * @brief SPI2 Initialization Function
  * @param None
  * @retval None
  */
static void MX_SPI2_Init(void)
{
  /* USER CODE BEGIN SPI2_Init 0 */
  /* USER CODE END SPI2_Init 0 */
  /* USER CODE BEGIN SPI2_Init 1 */
  /* USER CODE END SPI2_Init 1 */
  /* SPI2 parameter configuration*/
  hspi2.Instance = SPI2;
  hspi2.Init.Mode = SPI_MODE_MASTER;
  hspi2.Init.Direction = SPI_DIRECTION_2LINES;
  hspi2.Init.DataSize = SPI_DATASIZE_8BIT;
  hspi2.Init.CLKPolarity = SPI_POLARITY_HIGH;
```

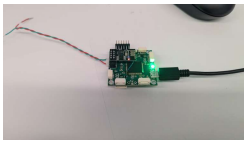
```
  hspi2.Init.CLKPhase = SPI_PHASE_2EDGE;
  hspi2.Init.NSS = SPI_NSS_SOFT;
  hspi2.Init.BaudRatePrescaler = SPI_BAUDRATEPRESCALER_64;
  hspi2.Init.FirstBit = SPI_FIRSTBIT_MSB;
  hspi2.Init.TIMode = SPI_TIMODE_DISABLE;
  hspi2.Init.CRCCalculation = SPI_CRCCALCULATION_DISABLE;
  hspi2.Init.CRCPolynomial = 10;
  if (HAL_SPI_Init(&hspi2) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN SPI2_Init 2 */
  /* USER CODE END SPI2_Init 2 */
}
/**
  * @brief GPIO Initialization Function
  * @param None
  * @retval None
  */
static void MX_GPIO_Init(void)
{
  GPIO_InitTypeDef GPIO_InitStruct = {0};
  /* USER CODE BEGIN MX_GPIO_Init_1 */
  /* USER CODE END MX_GPIO_Init_1 */
  /* GPIO Ports Clock Enable */
  __HAL_RCC_GPIOE_CLK_ENABLE();
  __HAL_RCC_GPIOC_CLK_ENABLE();
  __HAL_RCC_GPIOH_CLK_ENABLE();
  __HAL_RCC_GPIOA_CLK_ENABLE();
  __HAL_RCC_GPIOB_CLK_ENABLE();
  __HAL_RCC_GPIOD_CLK_ENABLE();
  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOE, VDD_3V3_SENSORS_EN_Pin|MAG_CS_Pin, GPIO_PIN_SET);
  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOC, ICM2_CS_Pin|ICM_CS_Pin|VDD_3V3_PERIPH_EN_Pin, GPIO_PIN_SET);
  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOB, GREEN_LED_Pin|RED_LED_Pin|BLUE_LED_Pin, GPIO_PIN_SET);
  /*Configure GPIO pin Output Level */
  HAL_GPIO_WritePin(GPIOD, FRAM_CS_Pin|BARO_CS_Pin, GPIO_PIN_SET);
  /*Configure GPIO pins : VDD_3V3_SENSORS_EN_Pin MAG_CS_Pin */
  GPIO_InitStruct.Pin = VDD_3V3_SENSORS_EN_Pin|MAG_CS_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOE, &GPIO_InitStruct);
  /*Configure GPIO pins : ICM2_CS_Pin ICM_CS_Pin VDD_3V3_PERIPH_EN_Pin */
  GPIO_InitStruct.Pin = ICM2_CS_Pin|ICM_CS_Pin|VDD_3V3_PERIPH_EN_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);
  /*Configure GPIO pins : GREEN_LED_Pin RED_LED_Pin BLUE_LED_Pin */
  GPIO_InitStruct.Pin = GREEN_LED_Pin|RED_LED_Pin|BLUE_LED_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);
  /*Configure GPIO pins : FRAM_CS_Pin BARO_CS_Pin */
  GPIO_InitStruct.Pin = FRAM_CS_Pin|BARO_CS_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(GPIOD, &GPIO_InitStruct);
  /* USER CODE BEGIN MX_GPIO_Init_2 */
  /* USER CODE END MX_GPIO_Init_2 */
}
/* USER CODE BEGIN 4 */
/* USER CODE END 4 */
/**
  * @brief  This function is executed in case of error occurrence.
  * @retval None
  */
void Error_Handler(void)
{
  /* USER CODE BEGIN Error_Handler_Debug */
    /* User can add his own implementation to report the HAL error return state */
    __disable_irq();
    while (1)
    {
    }
  /* USER CODE END Error_Handler_Debug */
}
#ifdef USE_FULL_ASSERT
/**
  * @brief  Reports the name of the source file and the source line number
  *         where the assert_param error has occurred.
  * @param  file: pointer to the source file name
  * @param  line: assert_param error line source number
  * @retval None
  */
void assert_failed(uint8_t *file, uint32_t line)
{
  /* USER CODE BEGIN 6 */
    /* User can add his own implementation to report the file name and line number,
       ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
  /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */
```

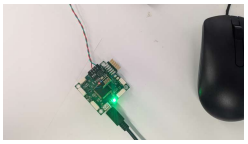
moving into esp testing, need to write/find simple code for the actual esps then check esp->stm connection then look at reading esp base station using Arduino or something idk

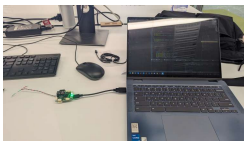
02/02/2026-

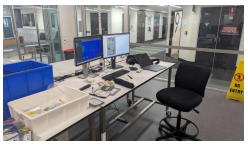
V2 board fabricated, lights are white so all key sensors com. Looks good very happi











28/01/2026-

I sent 12.5 Volts into board and it caught fire.

Had to replace all regulators, IC11 U1 U2. Seemed to fix got off lucky. Stupid ass ESC breakout didn't come with BEC and the V is 12 like wtf

02/02/2026-

Finished all fabrication of the board and moved into starting firmware programming. Need to edit opensource PX4 and add mavlink8266 to small wifi chip on board. Using my arch linux computer for this, need the px4 github repo|modify drivers and spi Chip select in board config), the mavlink8266 firmware repo. Lots of issues with this so take your time solving them if repeating.

03/02/2026-

Started with mavlink8266 firmware, I got the px4 firmware loaded but no sensors were reading so I stopped. I got a ESP-USB-SERIAL github repo and followed the instructions to turn a small esp3253 into a usb->serial converter, then used platform io and my computer to flash the esp-01_1M mavlink software onto the wifi chip. Was a pain but this worked well once setup(pain in the ass). Before that I spend time ripping apart a small racing drone I owned (same comps that send 12V into my board) to use as a small testing rig for external batt power and basic motor control. No props and need to upgrade but it was a good small platform for initial testing, no radio need to steal one from steve

09/02/2026-

Now looking into PX4 firmware, spent a lot of the day going WTF but using gemini realised that I had only added the new sensor drivers and the CS pins were not updated in firmware(this was mentioned above but im writing all of this now so hindsight's annoying). After modifying and re-flashing (MAKE SURE THE CHIP HAS THE BOOTLOADER INSTALLED OTHERWISE PX4 FLASH DON'T WORK) I got sensors to read, idk about accuracy or errors but we can fix that up later. Wanted wifi now since I did all that effort above, You set MAVLINK .0 to wifi port on QGroundControl then go to serial and set wifi serial to 912000bit then it all works perfectly(had to figure that out the hard way), noticed that imu data is not transferred wirelessly. Could control motor outputs on the board wirelessly from ground control which was a huge success, I have a video but that vid files would prob get lost. Moving into calibrating sensors correctly, connecting antenna so Qgroundcontrol errors would go away. Once they are all gone may speak to steve about drone flying using board(we will break a drone probs)

10/02/2026-

Spent this time preparing for the first flight, was given access to a large holy bro drone and added my board to the drone. Also tested mavlink wifi for the board and added a RC connection so the pilot had proper equipment for test. Couldn't test the props or battery reader without the pilot and safety net so after tests complete we were done for the day.

20/02/2026-

TESTDAY, Wrote up a design document slides for the pilot. These detailed the board specs and changes. At 1 we went to level 1 oblock and met the pilot (the goat)

Julian Galvez Serna

He configured the drone and ran some initial tests that standard pilots do. This was then looking at battery install and prop testing (M3 M4 had to be switched thank you Julian for the pickup). Then moved the drone into the netted area and preformed Maiden flight for the board, all tests were green and there was some sway in the idle hover(manual mode). Very happy !

Moving to adding a companion computer and uart telemety 1 connection so auto pilot mode can be tested(using inside camera tracking system to map the drones position and using these position values to tune the autopilot mode)

24/02/2026-

I have added a pi4 b with the custom power board. The Fmu board was attached above and a custom USB->uart converter was made using a esp32-S3 to break out into a JSTGH connector for TELM1. This was tested with a ros2 download and the mavros node. Ros2 was installed Using Tobias's ROBOSTACK approach and the result showed that ros2 could visualize the imu output. This proved that the FMU board could connect to a companion computer. Testing of this was to occur the next day

SECOND FLIGHT TEST

The second flight test tested sensor fusion using a indoor tracking system and a compantion computer. Due to the previous tests all that was need to be modified was the ACM0 and baudrate. The first issue was there was large error between the Indoor position readings and the Accelerometer position reeadings. This was fixed by turning on a EFK setting. The Height was then failing this was fixed by turning off the barometer and only using indoor position data. The drone had some issue with yaw drift however this could have been due to a poor weight distribution on the airframe. The first test was a position hold test using the indoor tracking system and that worked. When moving to the final waypoint test a error which stopped offboard mode from activating occurred every time during flight. No cause was identified and the test concluded. Waypoint mode would have worked ideally if offboarding mode could be activated mid flight, this also may have been linked to the RC mode switching not working aswell. These 2 firmware issues are the key issues to fix in code before the board can move on to next phase(accuracy definitions or something in that scope)

27/02/26-

Compiling files for handover end of this log as no new process being made